Student Enrollment & Course Management System

User Story:

A student logs into the system, searches for a course they are interested in, and enrolls after verifying prerequisites. The student receives a confirmation email and can view the course details and schedule in their dashboard.

1. Introduction

1.1 Project Overview

- This section outlines the purpose and scope of the Student Enrollment & Course Management System (SECMS).
- The system facilitates the seamless enrollment of students into courses, manages student data, and enables easy access for both students and administrators.

1.2 Purpose

- The primary goal of the project is to simplify and automate the student enrollment process and course management.
- This includes functionalities such as student registration, course selection, schedule management, and providing reports for academic tracking.

2. Ideation Phase

2.1 Problem Statement

The current manual system of course enrollment is inefficient and prone to errors.

Students face difficulties in managing their schedules, and administrators struggle to track enrollment and course progress.

2.2 Empathy Map Canvas

This tool helps us to empathize with the users (students and administrators), understand their needs, pain points, and motivations, which are essential for designing a user-friendly system.

2.3 Brainstorming & Idea Prioritization

- Step 1: Brainstorming Ideas
  - A collaborative session with stakeholders to generate ideas on how to improve the system.
- Step 2: Grouping Ideas

    - Categorizing the ideas based on priority and relevance.
- Step 3: Prioritization
    - Prioritizing ideas based on feasibility, impact, and ease of implementation.

3. Requirement Analysis Phase

3.1 Customer Journey Map

Mapping the entire experience a student and administrator would go through while using the system. This helps in understanding the flow and identifying potential issues.

3.2 Solution Requirements

Outlining the required functionalities, such as user registration, course management, payment processing, etc.

3.3 Data Flow Diagram

Visualizing how data will flow through the system from input (student details, course data) to output (student's course enrollment, academic performance).

3.4 Technology Stack

Identifying the technologies and tools that will be used to build the system, such as programming languages, frameworks, databases, and platforms.

4. Project Design Phase

4.1 Problem-Solution Fit

Ensuring that the solution addresses the problem effectively by aligning the features with the pain points identified in the ideation phase.

4.2 Proposed Solution

Presenting the system's architecture and features in detail, explaining how it will meet the requirements of both students and administrators.

4.3 Solution Architecture

A detailed breakdown of the system's architecture, including front-end, back-end, database, and third-party integrations.

Functional Requirements

Student Registration: Students must be able to register and create an account.

Course Enrollment: Students can browse available courses and enroll.

Grade Tracking: Students can track their academic performance in each enrolled course.

Admin Dashboard: Administrators can manage courses, enroll students, and track academic progress.

Payment Gateway: Integration with a payment system for fee payments.

Non-Functional Requirements

Usability: The system should be easy to use for both students and administrators.

Security: The system must ensure secure handling of student data, including personal and financial information.

Reliability: The system should function without crashes or errors.

Performance: The system should handle a large number of simultaneous users efficiently.

Scalability: The system should support adding more students, courses, and features as needed.

Availability: The system should be available and functional 24/7, with minimal downtime.

5. Project Planning & Scheduling

5.1 Project Planning

- Define a detailed roadmap of the project's development lifecycle.
- Identify project milestones, key deadlines, and deliverables.
- Define team roles and responsibilities.
- Set up communication protocols for updates, meetings, and reviews.

Product Backlog, Sprint Schedule, and Estimation

Product Backlog: List and prioritize the features, tasks, and user stories required for the system. The backlog will be divided into actionable tasks for each sprint.

Sprint Schedule: Create a timeline for each sprint. Each sprint should have a clear goal and deadline.

Estimation: Estimate the effort and time required for each task, considering the complexity and dependencies.

6. Project Development Phase

Project Flow

Design and Architecture:

- Develop a high-level system architecture, including both front-end and back-end components.
- Create wireframes and UI/UX designs based on user stories and feedback.
- Design database schemas and integrate necessary third-party systems (like payment gateways).

Development Phases:

Front-end Development: Develop the student-facing portal for course enrollment, profile management, notifications, etc.

Back-end Development: Build the administrative dashboard for course management, user registration, reports, and payment handling.

Integration: Integrate the front-end and back-end systems and ensure seamless data flow and user experience.

Testing and Quality Assurance:

Unit Testing: Test individual components like course enrollment, notifications, and payment systems.

Integration Testing: Ensure that all components work together as expected.

User Acceptance Testing (UAT): Allow users (students and administrators) to test the system and provide feedback.

Deployment:

Deploy the system to a live environment, ensuring that it is scalable, secure, and optimized for user performance.

7. Maintenance Phase

- After the system goes live, monitor performance and user feedback.
- Provide regular updates, bug fixes, and new features as needed.
- Ensure the system remains secure and up-to-date with technological advancements.

8. Risk Management

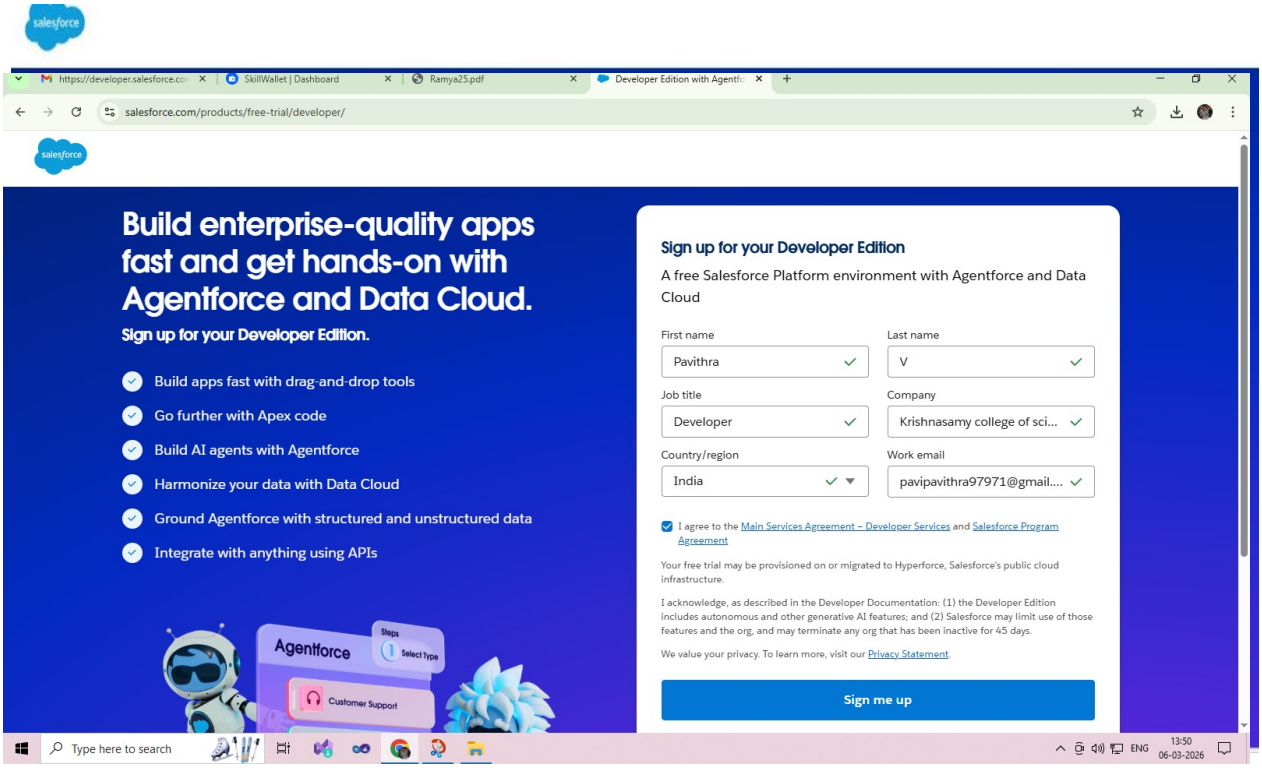
- Identify potential risks such as system downtime, data breaches, and scope creep.
- Develop mitigation strategies, including backup systems, security protocols, and regular reviews.

Project Flows:

Milestone 1: Salesforce Developer Account Creation

Objective: Set up a Salesforce Developer account to begin developing the system.

Go To: https://www.salesforce.com/products/free-trial/developer/



Tasks:

- Visit the Salesforce Developer website and sign up for a free account.
- Set up the development environment in Salesforce, including accessing the Salesforce Developer Console and Salesforce Setup menu.
- Familiarize yourself with the Salesforce Lightning platform.

Milestone 2: Custom Object Creation (Student, Course, Enrollment, Instructor)

Objective: Create the foundational custom objects for the system.

Tasks:

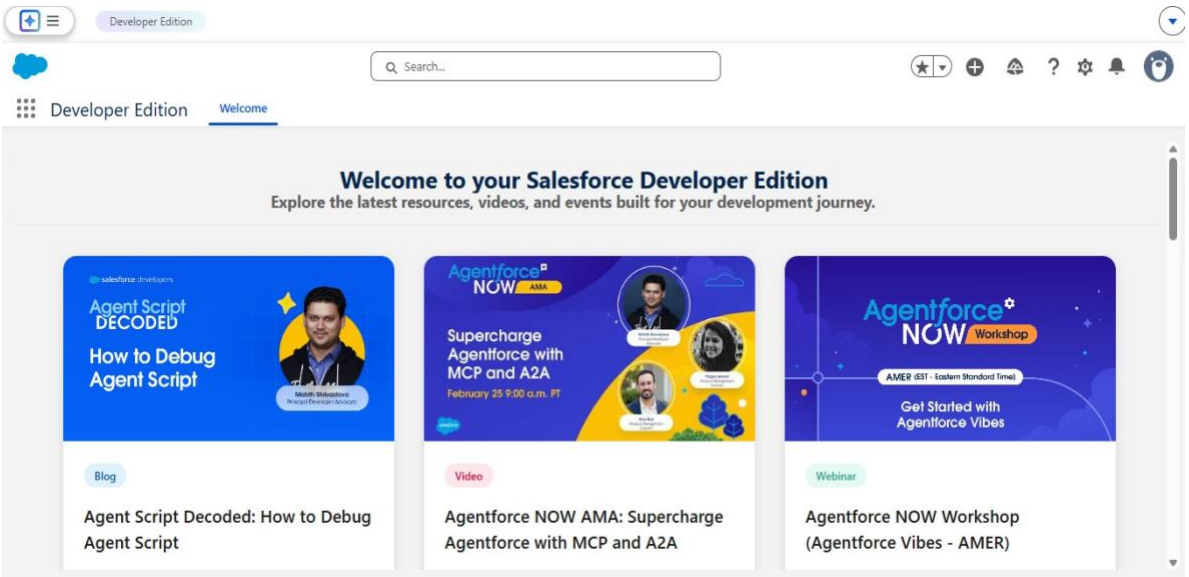
Student Object: Create a custom object to store student information (name, contact details, academic year, etc.).

Course Object: Create a custom object to store course details (course name, description, duration, prerequisites).

Enrollment Object: Create an object to link students with courses (student ID, course ID, enrollment date).

Instructor Object: Create an object for instructors (name, department, courses taught).

Ensure each custom object has relevant fields and validation rules to store necessary data.



Milestone 3: Tabs Creation

Objective: Create tabs for easy access to the custom objects within the Salesforce platform.

Tasks:

- Create custom tabs for each object (Student, Course, Enrollment, Instructor).
- Customize the layout of each tab to display the relevant fields and records.
- Test the tab navigation to ensure smooth access to all the objects for both administrators and students.

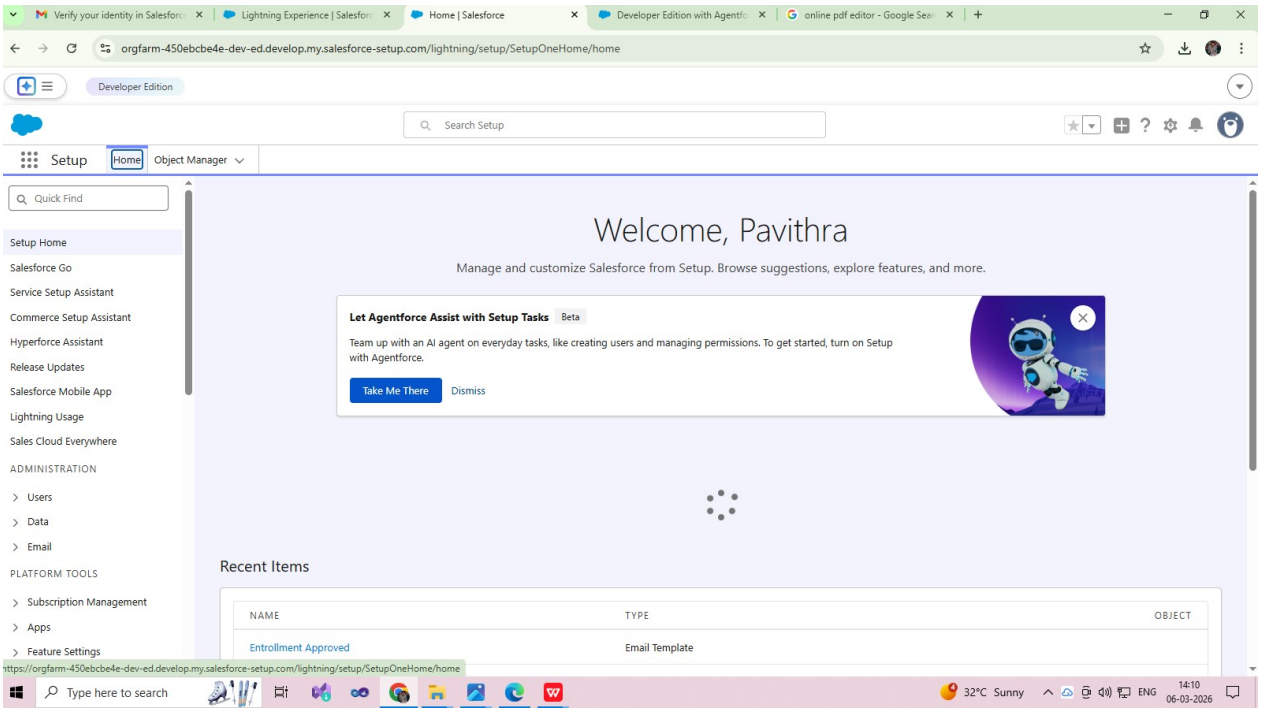
Milestone 4: Lightning App Creation

Objective: Set up a Salesforce Lightning App that provides an intuitive user interface.

Tasks:

- Create a Lightning App to consolidate the various tabs (Student, Course, Enrollment, Instructor) into one user-friendly interface.

- Customize the app's navigation and layout based on user needs (e.g., students, administrators).
- Add functionality like the ability to view student records, enroll in courses, and manage instructor details from within the app.



Milestone 5: Fields Creation

Objective: Set up fields for each custom object to capture the necessary data.

Tasks:

- Create fields in the Student object for data such as student name, email, phone number, academic year, etc.
- In the Course object, create fields for course name, description, duration, and prerequisites.
- In the Enrollment object, include fields like enrollment date, student ID, course ID, status, etc.
- In the Instructor object, create fields for instructor name, department, courses taught, etc.
- Define field types (text, date, picklist, etc.) based on the data requirements.

Milestone 6: Create Relationships

Objective: Define the relationships between the custom objects to ensure data consistency.

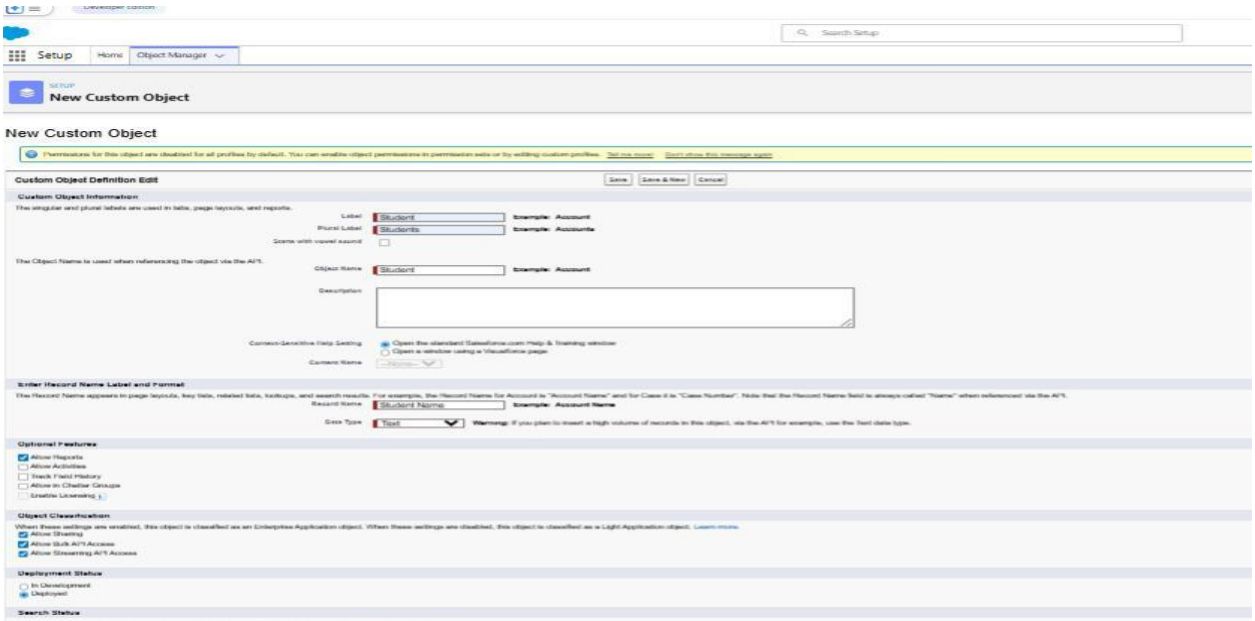
Tasks:

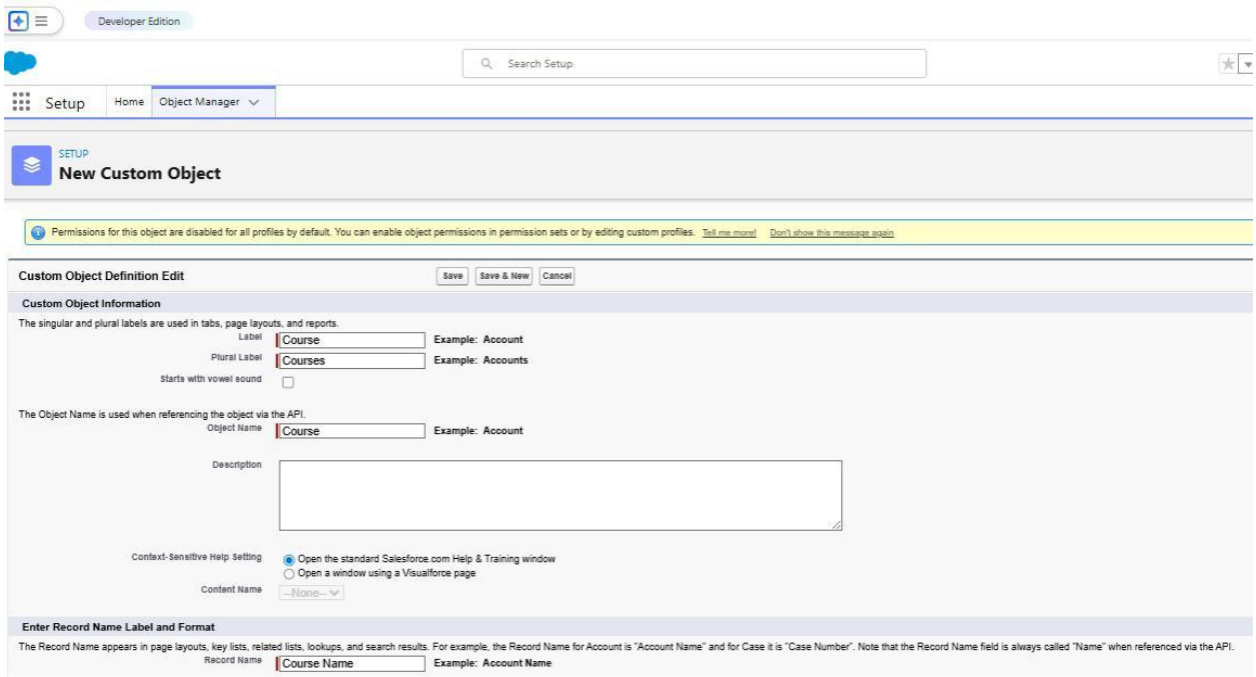
Student to Enrollment: Create a Master-Detail Relationship between the Student object and the Enrollment object to represent which students are enrolled in which courses.

Course to Enrollment: Create a Master-Detail Relationship between the Course object and the Enrollment object to capture which courses a student is enrolled in.

Instructor to Course: Create a Lookup Relationship between the Instructor object and the Course object, indicating which instructor is teaching which course.

Review and test the relationships to ensure data integrity (e.g., enrolling a student in a course should automatically link to the correct course and instructor).





Milestone 7: Page Layout Customization

Objective: Customize the page layouts for each object to ensure relevant fields and components are displayed for users.

Tasks:

Student Object Layout: Customize the layout to display the most relevant information for students, including personal details, enrolled courses, and contact information.

Course Object Layout: Customize the layout to display course details such as name, description, prerequisites, and enrolled students.

Enrollment Object Layout: Customize this layout to display student enrollment information, status, and course details.

Instructor Object Layout: Adjust the layout to display instructor details, including name, department, courses taught, and contact information.

Use Related Lists to link objects and display relationships between them (e.g., show a student's enrolled courses on the student record page).

Milestone 8: Validation Rules

Objective: Set up validation rules to ensure that the data entered is correct and follows the required format.

Tasks:

Student Object Validation: Ensure that required fields like student name, email, and phone number are entered correctly. Implement rules to validate email formats and phone number formats.

Course Object Validation: Set validation rules to ensure course details are entered correctly (e.g., course name should not be empty, duration should be a positive number).

Enrollment Object Validation: Validate that students cannot be enrolled in the same course multiple times, and that enrollment dates are valid.

Instructor Object Validation: Ensure that instructors are assigned to the correct course and department.

Milestone 9: Flows (Automation)

Objective: Set up Salesforce Flows for automating business processes and enhancing system functionality.

Tasks:

Enrollment Flow: Automate the enrollment process, including sending email notifications to students after successful enrollment and confirming their schedule.

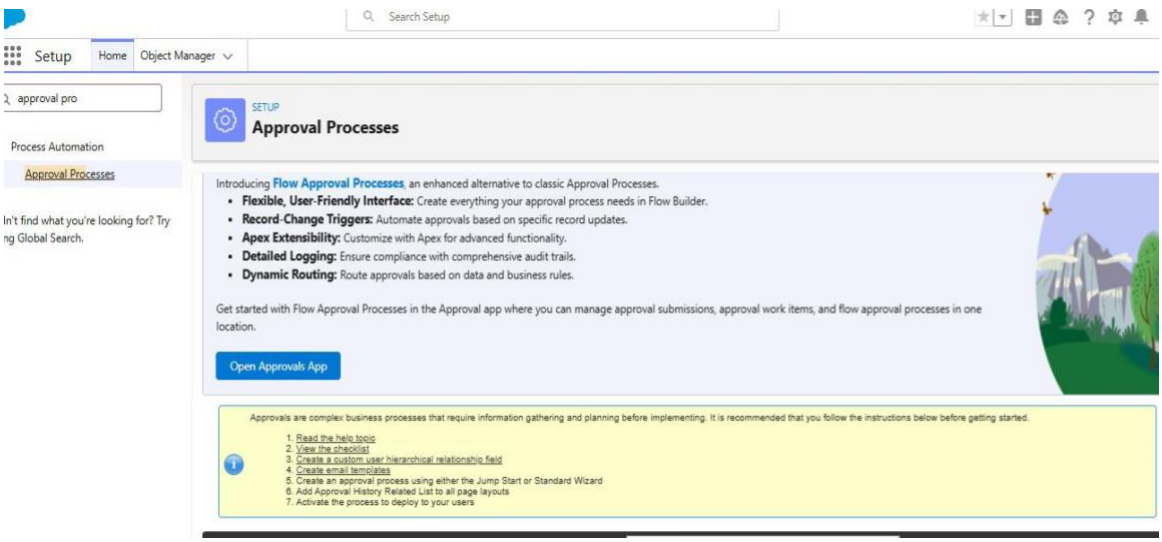
Student Registration Flow: Automate the process of registering new students, sending them a welcome email, and linking their records to the appropriate courses.

Instructor Assignment Flow: Automatically assign instructors to courses based on course requirements and availability.

Course Notification Flow: Set up a flow to automatically notify students about important course updates, such as schedule changes or exam dates.

Milestone 10: Approval Process

Objective: Implement approval processes for certain tasks that need managerial oversight (e.g., course approvals, student enrollment requests).



Tasks:

Course Approval Process: Set up an approval process where new course creation or changes must be approved by an administrator or department head before they are made live.

Enrollment Approval Process: Configure an approval process for course enrollments where certain courses or student enrollments need manager approval based on criteria such as prerequisites or class capacity.

Set up approval stages, notifications, and automated actions after the approval process is completed.

Milestone 11: Reports

Objective: Create custom reports to analyze and track system data (e.g., student enrollments, course statistics).

Tasks:

Student Enrollment Report: Create a report that shows the number of students enrolled in each course, including details like course name, student name, and enrollment date.

Course Statistics Report: Generate a report to track course popularity, number of students enrolled, and instructor performance.

Instructor Report: Create a report showing which courses each instructor is teaching, their availability, and the number of students in each course.

Custom Reports: Develop additional reports for administrators to track key metrics and improve decision-making.

Milestone 12: Dashboards

Objective: Create dashboards to visualize key data and make it easier for users to access important information at a glance.

Tasks:

Student Dashboard: Create a dashboard for students that shows their enrolled courses, upcoming exams, and performance metrics.

Administrator Dashboard: Develop a comprehensive dashboard for administrators to track overall course enrollment, student performance, and approval processes.

Instructor Dashboard: Create a dashboard that displays a teacher's course roster, student performance, and upcoming tasks.

Ensure that the dashboards are dynamic, with real-time data and customizable views.

Milestone 13: Security Setup

Objective: Configure security settings to protect data, define user access, and ensure the system complies with privacy standards.

Tasks:

User Profiles: Create and configure profiles for students, instructors, and administrators with appropriate permissions (e.g., access to specific records, reports, and dashboards).

Role Hierarchy: Set up role hierarchies to ensure that users at different levels have the correct access to information.

Field-Level Security: Restrict access to sensitive fields such as student contact information or course fees, ensuring only authorized users can view or edit them.

Sharing Settings: Set up sharing rules to define how records (students, courses, enrollments) are shared within the organization based on roles and criteria.

Two-Factor Authentication (2FA): Enable two-factor authentication for admins and other users who need elevated security levels.

Advantages of SECMS

Efficiency:

- Automates the student enrollment and course management processes, reducing the time and effort spent on manual tasks.
- Students can quickly enroll in courses online, and administrators can easily track and manage enrollments.

User-Friendly Interface:

- The system provides an intuitive interface for students, instructors, and administrators to navigate easily.
- Lightning apps and customized page layouts make it easy for each user to access relevant data.

Real-Time Updates:

- Students and administrators receive real-time updates on course enrollment, class schedules, and academic performance.
- Automated email notifications keep users informed about key changes and events.

Data Accuracy:

- Validation rules ensure that data entered into the system is accurate, reducing errors and inconsistencies.
- Real-time data analysis helps maintain up-to-date information for reports and decision-making.

Security:

- Strong security measures such as role-based access and field-level security ensure that sensitive information, such as student contact details and grades, is protected.

Scalability:

- The system is designed to handle large numbers of students, courses, and instructors, making it scalable as the institution grows.

Disadvantages of SECMS:

Initial Setup Complexity:

- The development and configuration of custom objects, relationships, and automation features in Salesforce can be complex for users unfamiliar with the platform.
- A significant amount of time may be required to set up the system properly and customize it according to the institution's needs.

Dependence on Salesforce Platform:

- The system relies heavily on the Salesforce platform, meaning any disruptions in Salesforce services (e.g., outages or updates) could potentially affect the system's availability.
- Additional costs may be incurred for Salesforce subscriptions or integrations with third-party tools.

Conclusion:

- **Objective:** Finalize the project, conduct a review, and prepare for deployment and future enhancements.

Tasks:

- **Testing:** Conduct final testing of the system, ensuring all features (enrollment, reporting, approval processes, etc.) are functioning as expected.
- **User Training:** Provide training to the system users (students, instructors, administrators) on how to use the system effectively.
- **Deployment:** Deploy the system to the production environment and ensure that all data is migrated correctly.
- **Documentation:** Finalize the project documentation, including user manuals, developer guides, and deployment instructions.
- **Future Enhancements:** Identify areas for future enhancements such as mobile app integration, additional reporting features, and advanced analytics.

Future Scope :

Mobile App Integration:

- The SECMS can be expanded into a mobile app, allowing students and instructors to access the system on the go, manage enrollments, and view course details from their smartphones.

AI-Powered Features:

- Implement machine learning algorithms to recommend courses to students based on their previous enrollments and academic interests.
- Integrate AI to help instructors in grading assignments and providing personalized feedback to students.

Advanced Reporting and Analytics:

- Add more sophisticated reporting tools, including predictive analytics, to help administrators forecast student performance, course demand, and instructor efficiency.
- Provide customizable dashboard options for more granular insights.

Integration with External Systems:

- Future integrations could include linking SECMS with external learning management systems (LMS), payment gateways, or academic performance tracking tools for more comprehensive functionality.

Multi-Language and Multi-Currency Support:

- Expand the system to support multiple languages for use in international institutions, and provide multi-currency payment options for fee payments.

Appendix

System Requirements:

- Hardware: Recommended server configurations for hosting the Salesforce instance (if necessary).
- Software: List of tools and technologies used, including Salesforce version, programming languages, and integrations.

Code Snippets:

- Include relevant code for setting up Flows, custom objects, validation rules, or any automation implemented in the system.

User Manuals:

- Provide a guide for administrators and end-users detailing how to navigate the system, enroll students, manage courses, and generate reports.